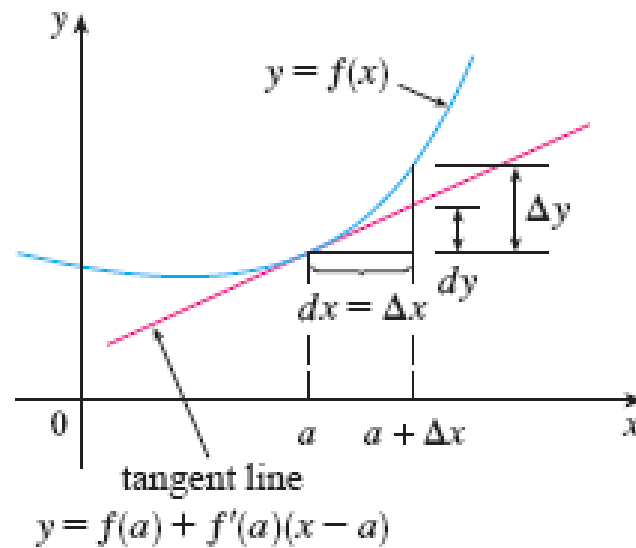
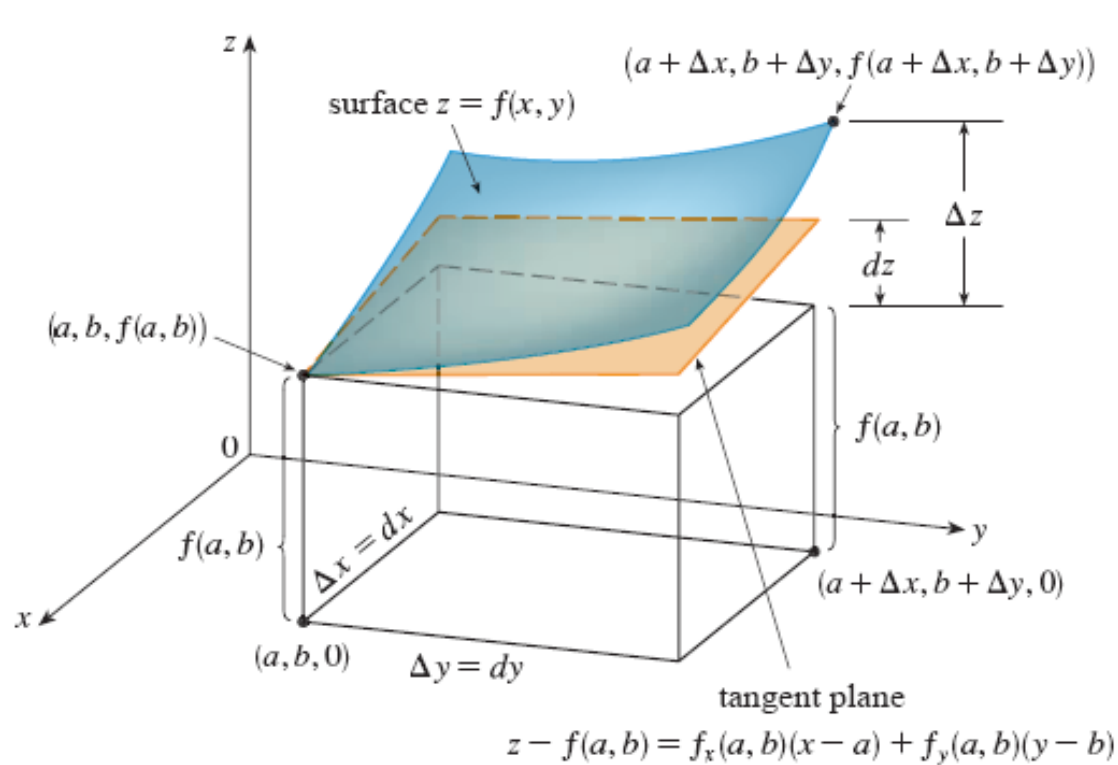


# Differential: Function of one variable



# Differential: Function of two variables



$$dz = f_x(x, y) dx + f_y(x, y) dy = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy$$

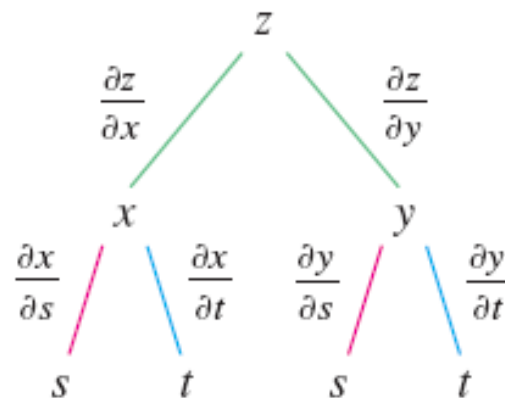
# The Chain Rule

**2 The Chain Rule (Case 1)** Suppose that  $z = f(x, y)$  is a differentiable function of  $x$  and  $y$ , where  $x = g(t)$  and  $y = h(t)$  are both differentiable functions of  $t$ . Then  $z$  is a differentiable function of  $t$  and

$$\frac{dz}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}$$

**3 The Chain Rule (Case 2)** Suppose that  $z = f(x, y)$  is a differentiable function of  $x$  and  $y$ , where  $x = g(s, t)$  and  $y = h(s, t)$  are differentiable functions of  $s$  and  $t$ . Then

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial s} \qquad \frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial t}$$



**4 The Chain Rule (General Version)** Suppose that  $u$  is a differentiable function of the  $n$  variables  $x_1, x_2, \dots, x_n$  and each  $x_j$  is a differentiable function of the  $m$  variables  $t_1, t_2, \dots, t_m$ . Then  $u$  is a function of  $t_1, t_2, \dots, t_m$  and

$$\frac{\partial u}{\partial t_i} = \frac{\partial u}{\partial x_1} \frac{\partial x_1}{\partial t_i} + \frac{\partial u}{\partial x_2} \frac{\partial x_2}{\partial t_i} + \dots + \frac{\partial u}{\partial x_n} \frac{\partial x_n}{\partial t_i}$$

for each  $i = 1, 2, \dots, m$ .



1. If  $z = x^2y + 3xy^4$ , where  $x = \sin 2t$  and  $y = \cos t$ , find  $\frac{dz}{dt}$  when  $t = 0$ .

2. If  $z = e^x \sin y$ , where  $x = st^2$  and  $y = s^2t$ , find  $\frac{\partial z}{\partial s}$  and  $\frac{\partial z}{\partial t}$ .

3. If  $u = x^4y + y^2z^3$ , where  $x = rse^t$ ,  $y = rs^2e^{-t}$ , and  $z = r^2s \sin t$ , find the value of  $\frac{\partial u}{\partial s}$  when  $r = 2$ ,  $s = 1$  and  $t = 0$ .

4. If  $g(s, t) = f(s^2 - t^2, t^2 - s^2)$  and  $f$  is differentiable, show that  $g$  satisfies the equation

$$t \frac{\partial g}{\partial s} + s \frac{\partial g}{\partial t} = 0.$$

5. If  $z = f(x, y)$  has continuous second order partial derivatives and  $x = r^2 + s^2$  and  $y = 2rs$ , find (a)  $\frac{\partial z}{\partial r}$  and (b)  $\frac{\partial^2 z}{\partial r^2}$ .



# Implicit Differentiation

The Chain Rule can be used to give a more complete description of the process of implicit differentiation that was introduced in Sections 3.5 and 14.3. We suppose that an equation of the form  $F(x, y) = 0$  defines  $y$  implicitly as a differentiable function of  $x$ , that is,  $y = f(x)$ , where  $F(x, f(x)) = 0$  for all  $x$  in the domain of  $f$ . If  $F$  is differentiable, we can apply Case 1 of the Chain Rule to differentiate both sides of the equation  $F(x, y) = 0$  with respect to  $x$ . Since both  $x$  and  $y$  are functions of  $x$ , we obtain

$$\frac{\partial F}{\partial x} \frac{dx}{dx} + \frac{\partial F}{\partial y} \frac{dy}{dx} = 0$$

But  $dx/dx = 1$ , so if  $\partial F/\partial y \neq 0$  we solve for  $dy/dx$  and obtain

6

$$\frac{dy}{dx} = -\frac{\frac{\partial F}{\partial x}}{\frac{\partial F}{\partial y}} = -\frac{F_x}{F_y}$$

To derive this equation we assumed that  $F(x, y) = 0$  defines  $y$  implicitly as a function of  $x$ . The **Implicit Function Theorem**, proved in advanced calculus, gives conditions under which this assumption is valid: it states that if  $F$  is defined on a disk containing  $(a, b)$ , where  $F(a, b) = 0$ ,  $F_y(a, b) \neq 0$ , and  $F_x$  and  $F_y$  are continuous on the disk, then the equation  $F(x, y) = 0$  defines  $y$  as a function of  $x$  near the point  $(a, b)$  and the derivative of this function is given by Equation 6.

A version of the **Implicit Function Theorem**

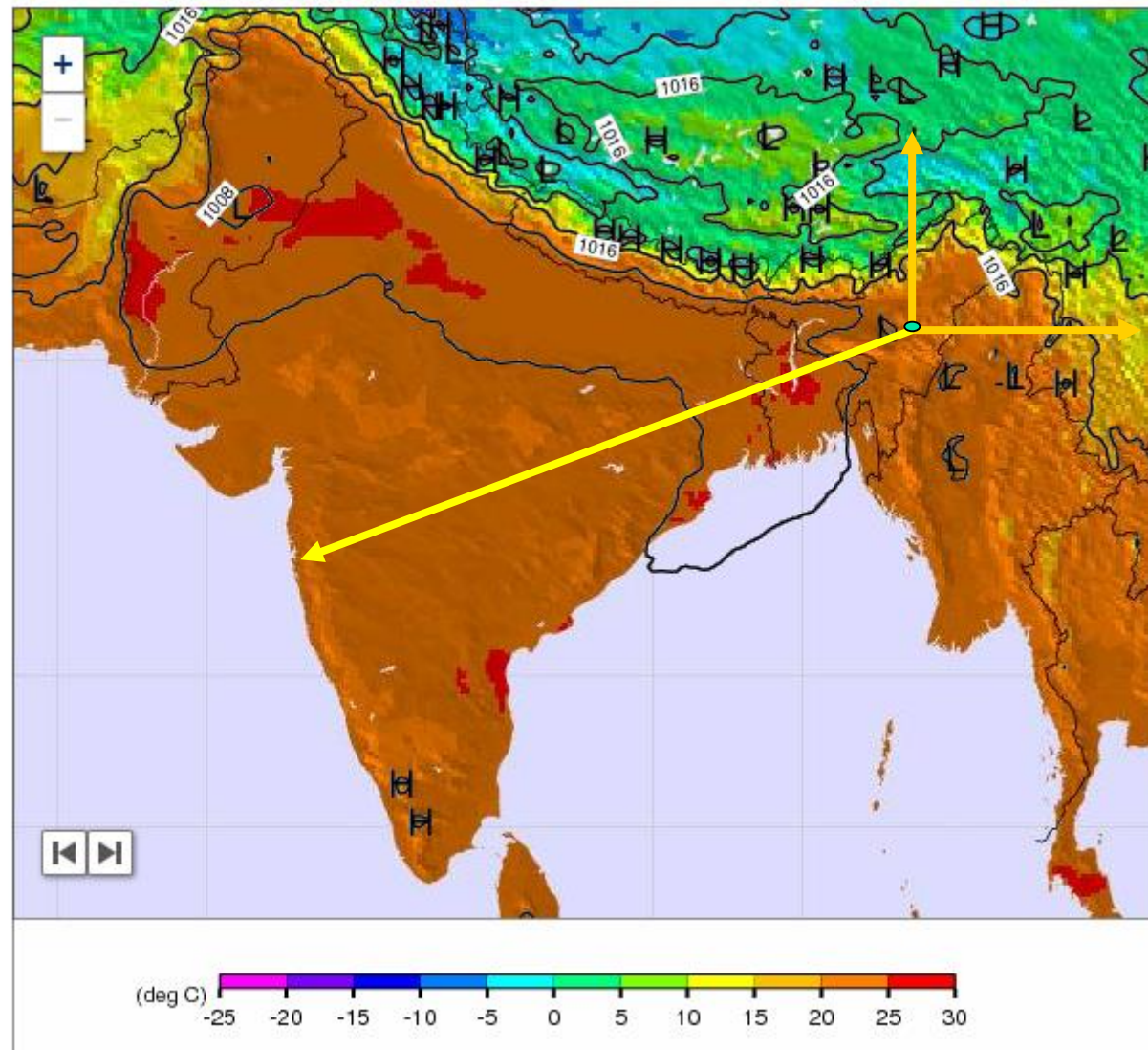
If  $F$  is defined within a sphere containing  $(a, b, c)$ , where  $F(a, b, c) = 0$ ,  $F_z(a, b, c) \neq 0$ , and  $F_x$ ,  $F_y$ , and  $F_z$  are continuous inside the sphere, then the equation  $F(x, y, z) = 0$  defines  $z$  as a function of  $x$  and  $y$  near the point  $(a, b, c)$  and this function is differentiable, with partial derivatives given by (7).

7

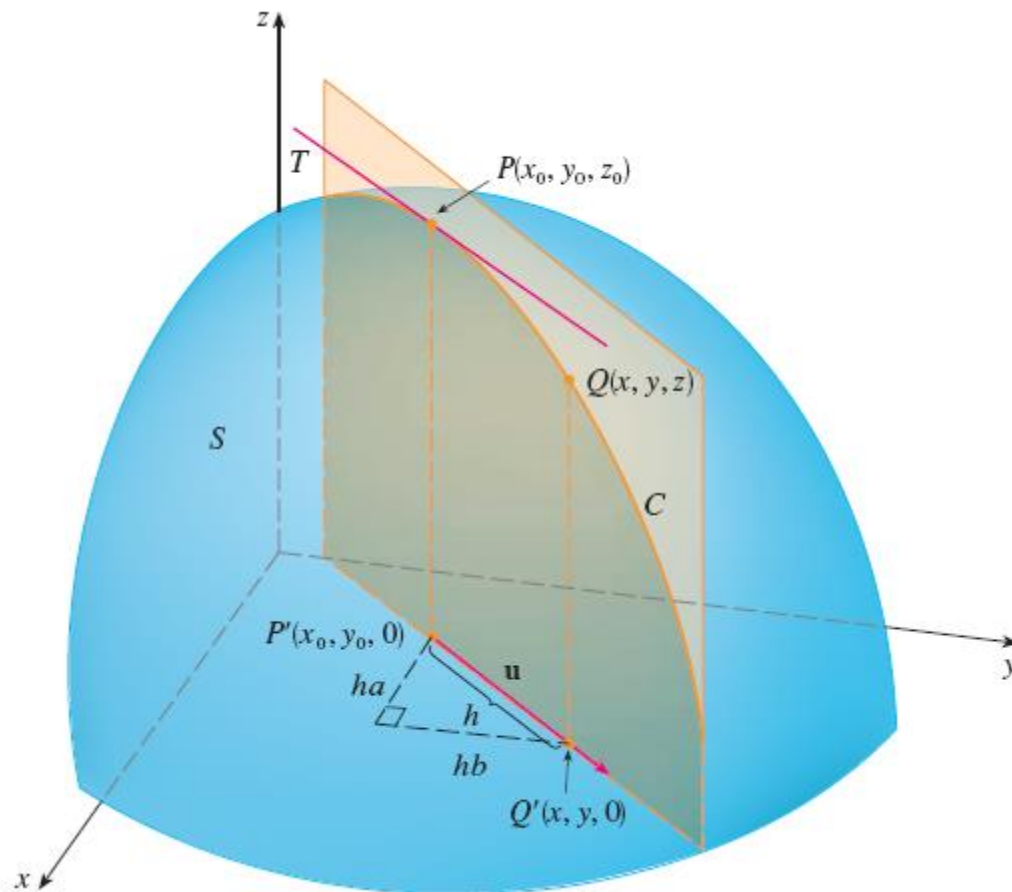
$$\frac{\partial z}{\partial x} = -\frac{\frac{\partial F}{\partial x}}{\frac{\partial F}{\partial z}} \quad \frac{\partial z}{\partial y} = -\frac{\frac{\partial F}{\partial y}}{\frac{\partial F}{\partial z}}$$

# DIRECTIONAL DERIVATIVE

Surface Temperature on Monday 09 Oct at 8:30am IST



9 and 10 October 2023

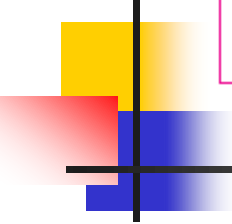


**2 Definition** The **directional derivative** of  $f$  at  $(x_0, y_0)$  in the direction of a unit vector  $\mathbf{u} = \langle a, b \rangle$  is

$$D_{\mathbf{u}} f(x_0, y_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + ha, y_0 + hb) - f(x_0, y_0)}{h}$$

if this limit exists.





**3 Theorem** If  $f$  is a differentiable function of  $x$  and  $y$ , then  $f$  has a directional derivative in the direction of any unit vector  $\mathbf{u} = \langle a, b \rangle$  and

$$D_{\mathbf{u}} f(x, y) = f_x(x, y) a + f_y(x, y) b$$

**EXAMPLE 2** Find the directional derivative  $D_{\mathbf{u}} f(x, y)$  if

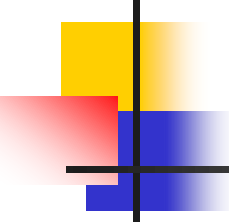
$$f(x, y) = x^3 - 3xy + 4y^2$$

and  $\mathbf{u}$  is the unit vector given by angle  $\theta = \pi/6$ . What is  $D_{\mathbf{u}} f(1, 2)$ ?

**8 Definition** If  $f$  is a function of two variables  $x$  and  $y$ , then the **gradient** of  $f$  is the vector function  $\nabla f$  defined by

$$\nabla f(x, y) = \langle f_x(x, y), f_y(x, y) \rangle = \frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j}$$

$$D_{\mathbf{u}} f(x, y) = \nabla f(x, y) \cdot \mathbf{u}$$



**10 Definition** The **directional derivative** of  $f$  at  $(x_0, y_0, z_0)$  in the direction of a unit vector  $\mathbf{u} = \langle a, b, c \rangle$  is

$$D_{\mathbf{u}} f(x_0, y_0, z_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + ha, y_0 + hb, z_0 + hc) - f(x_0, y_0, z_0)}{h}$$

if this limit exists.

If we use vector notation, then we can write both definitions (2 and 10) of the directional derivative in the compact form

**11**

$$D_{\mathbf{u}} f(\mathbf{x}_0) = \lim_{h \rightarrow 0} \frac{f(\mathbf{x}_0 + h\mathbf{u}) - f(\mathbf{x}_0)}{h}$$

where  $\mathbf{x}_0 = \langle x_0, y_0 \rangle$  if  $n = 2$  and  $\mathbf{x}_0 = \langle x_0, y_0, z_0 \rangle$  if  $n = 3$ . This is reasonable because the vector equation of the line through  $\mathbf{x}_0$  in the direction of the vector  $\mathbf{u}$  is given by  $\mathbf{x} = \mathbf{x}_0 + t\mathbf{u}$  (Equation 12.5.1) and so  $f(\mathbf{x}_0 + h\mathbf{u})$  represents the value of  $f$  at a point on this line.

Show that the operation of taking the gradient of a function has the given property. Assume that  $u$  and  $v$  are differentiable functions of  $x$  and  $y$  and that  $a, b$  are constants.

$$(a) \nabla(au + bv) = a \nabla u + b \nabla v \quad (b) \nabla(uv) = u \nabla v + v \nabla u$$

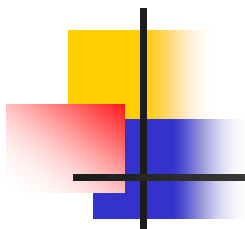
$$(c) \nabla\left(\frac{u}{v}\right) = \frac{v \nabla u - u \nabla v}{v^2} \quad (d) \nabla u^n = nu^{n-1} \nabla u$$

**V EXAMPLE 4** Find the directional derivative of the function  $f(x, y) = x^2y^3 - 4y$  at the point  $(2, -1)$  in the direction of the vector  $\mathbf{v} = 2\mathbf{i} + 5\mathbf{j}$ .

Show that if  $z = f(x, y)$  is differentiable at  $\mathbf{x}_0 = \langle x_0, y_0 \rangle$ , then

$$\lim_{\mathbf{x} \rightarrow \mathbf{x}_0} \frac{f(\mathbf{x}) - f(\mathbf{x}_0) - \nabla f(\mathbf{x}_0) \cdot (\mathbf{x} - \mathbf{x}_0)}{|\mathbf{x} - \mathbf{x}_0|} = 0$$

## Maximizing the directional derivative



**Suppose we have a function  $f$  of two or three variables and we consider all possible directional derivatives of  $f$  at a given point. These give the rates of change of  $f$  in all possible directions.**

**Q1:** In which of these directions does  $f$  change fastest?

**Q2:** What is the maximum rate of change?

**15 Theorem** Suppose  $f$  is a differentiable function of two or three variables. The maximum value of the directional derivative  $D_{\mathbf{u}} f(\mathbf{x})$  is  $|\nabla f(\mathbf{x})|$  and it occurs when  $\mathbf{u}$  has the same direction as the gradient vector  $\nabla f(\mathbf{x})$ .

# Tangent planes to level surfaces

16

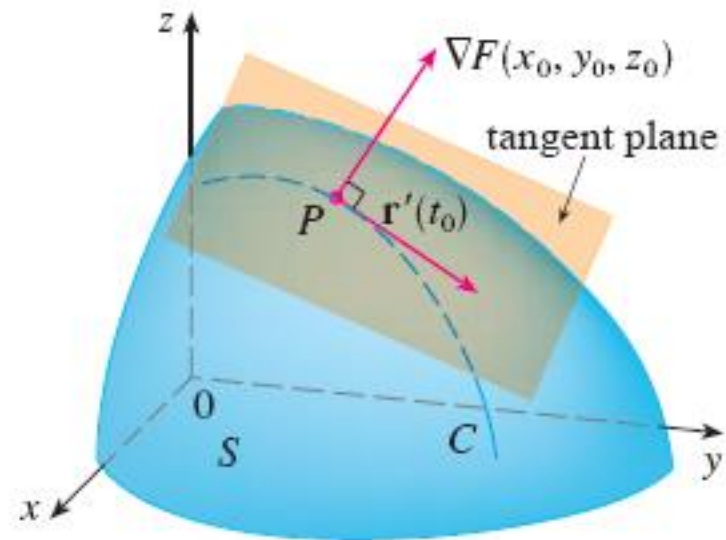
$$F(x(t), y(t), z(t)) = k$$

18

$$\nabla F(x_0, y_0, z_0) \cdot \mathbf{r}'(t_0) = 0$$

19

$$F_x(x_0, y_0, z_0)(x - x_0) + F_y(x_0, y_0, z_0)(y - y_0) + F_z(x_0, y_0, z_0)(z - z_0) = 0$$



What is its equivalent for the surface  $z=f(x, y)$ ?

The **normal line** to  $S$  at  $P$  is the line passing through  $P$  and perpendicular to the tangent plane. The direction of the normal line is therefore given by the gradient vector  $\nabla F(x_0, y_0, z_0)$  and its symmetric equations are

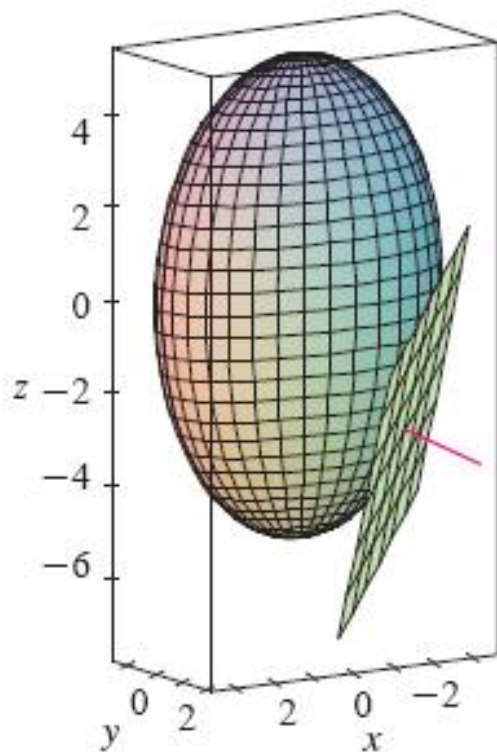
20

$$\frac{x - x_0}{F_x(x_0, y_0, z_0)} = \frac{y - y_0}{F_y(x_0, y_0, z_0)} = \frac{z - z_0}{F_z(x_0, y_0, z_0)}$$

**Example:** Show that  $\nabla\phi$  is a vector normal to the surface  $\phi(x, y, z) = c$ .

The significance of the above example is that if you are seeking a normal to a surface at a point, you have to simply find out the gradient vector of the surface at that point.

## Example 1



**V EXAMPLE 8** Find the equations of the tangent plane and normal line at the point  $(-2, 1, -3)$  to the ellipsoid

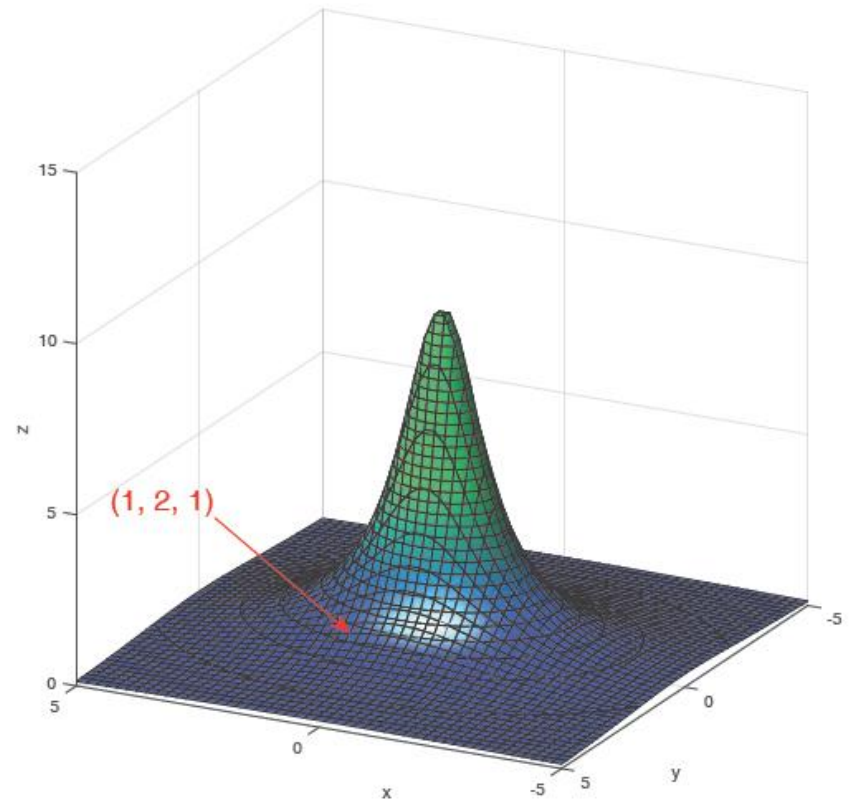
$$\frac{x^2}{4} + y^2 + \frac{z^2}{9} = 3$$

Suppose you are climbing a hill whose shape is given by the equation  $z = \frac{10}{1+x^2+2y^2}$

And you are standing at a point with coordinates  $(1, 2, 1)$ .

- In which direction should you proceed initially in order to reach the top of the hill fastest?

- If you climb in that direction, at what angle above the horizon will you be climbing initially?



# Significance of the gradient vector

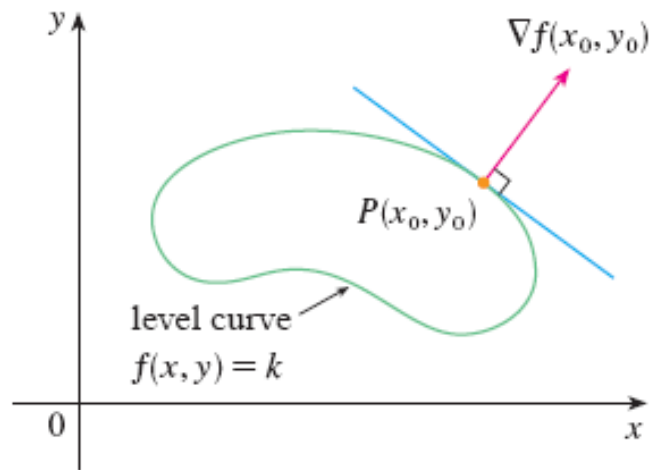


FIGURE 11

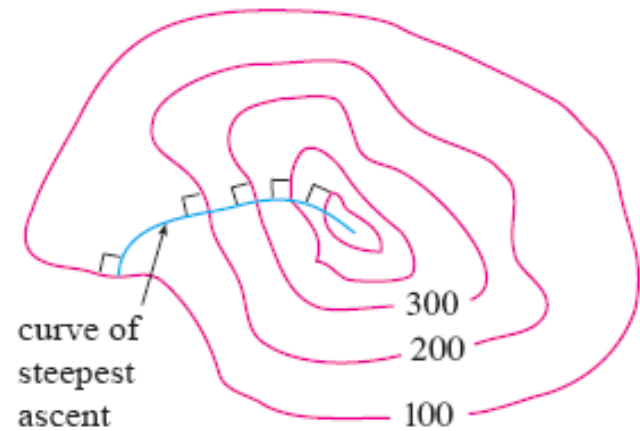


FIGURE 12